

Ethical Considerations in the Development of Low-Resource African Language NLP/LLMs

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ABSTRACT

Large Language Models (LLMs), notable for their ability to achieve general purpose NLP tasks, have advanced rapidly in recent years. LLMs have proved useful across many applications such as health and education. At the same time, their development has mostly been limited to a small number of the world's most digitized, and financially backed languages. This has left a large number of the world's languages underrepresented in NLP/LLM research, not least of which are the languages spoken in Sub-Saharan Africa. Small groups of researchers have begun to determine how best to develop NLP models for these data-scarce and less-backed languages. Our paper focuses on the ethical considerations in developing LLMs for the diverse range of African Languages which often lack substantial digital resources. We propose guidelines for creating culturally relevant and effective LLMs which promote equitable representation.

INTRODUCTION

The digital world is dominated by well-resourced languages. This means that language represented in and therefore supported by the internet are often backed by powerful political and financial communities. The concentrated development of ICT technologies like the Internet in the United States, Europe, and East Asia, has meant that languages not widely spoken in those regions are not adequately represented on the internet. Other historical and economic factors have also created a situation in which research and development of

ICT technologies by global-south-domiciled researchers is sparse.

One such research area is artificial intelligence, and more specifically Natural Language Processing (NLP) and Large Language Models (LLMs). These technologies promise to revolutionize education, business, healthcare, entertainment, art, and increase the accessibility of information. However, the benefits of these technologies are unevenly distributed, and threaten to become even more so. AI, and thus, LLMs, are often trained by scraping data from the internet. As discussed, Languages not represented well on the internet are also often spoken and used in regions which do not have the political or economic means to change that reality. The introduction of LLMs means that these language communities find themselves in a double-bind of having been excluded from the internet, and now being excluded from rapid NLP developments because the data it is trained on (the internet) is not readily available in their spoken or written languages. This is particularly true in Africa. Of the over 3,000 languages spoken in Africa, only a few, including European languages and Arabic, are adequately represented in the development of LLMs. This lack of representation presents a crisis for African populations as they risk being shut out of the benefits of NLP advancements. The lack of representation also poses a major threat to the cultural preservation of these African languages.

Conferences, research teams, organizations, and individuals have begun to take up the task of correcting the representational gap. These include *Lelape AI*, *Masakhane*, and more. Their activities include building effective NLP tools for African languages, but also building a scholarly community focused on the future of AI in Africa. The introduction of AI to Africa and African languages is in its nascent stages, and many researchers are seeking solutions for the aforementioned low-resource double-bind, which stalls the development of African language NLP tools.

Our research is motivated by the need to analyze these solutions and establish best practices in developing LLMs for the various low-resource African languages.

We ask 3 main questions. (1) What data collection and model training techniques are utilized in the development of low-resource African Large Language Models? (2) What are the ethical implications of these techniques as it relates to performance and cultural sensitivity? (3) What are the tradeoffs made when deciding on which techniques to use.

This paper explores adaptation techniques such as zero-shot learning and language adapters, assessing their efficacy and ethical implications. Our approach involved a literature review as well as interviews with leading experts in the field. Through these methods, we identified which training techniques were most promising and interrogated how different training techniques can or cannot be used together, and how their usage stacks up to ethical standards we gathered during the literature review. Our key findings reveal that while current models fall short in accurately representing the cultural specificities of many African Languages in translation, there are

options for improving NLP applications in translation, text, and speech.

From these insights, we organized a guide/table of training techniques which can help researchers better consider their options to develop more ethical and effective African Large Language Models. By prioritizing ethical considerations such as representation, accuracy, and cultural sensitivity, we can pave a way for the future where no language is left behind.

RELATED WORK

In recent years and months, scholarly literature on LLMs has begun to be published which focuses on representation, geographic bias, and low-resource languages. A common theme among these papers is the need for greater inclusivity and diversity in natural language processing technology. Mani and Namomsa (2023) explore the representational issues in large language models (LLMs) and stress the importance of addressing the underuse of multi-modal architecture for low-resource languages. Manvi et al. (2024) and Ojo et al. (2023) further emphasize these points by discussing the geographic bias in LLMs and their underperformance on African languages, respectively. Vukosi Marivate’s work on augmenting data for African NLP and the efforts by Zindi and AI4D in building language datasets specifically address these concerns by developing resources aimed at enhancing LLM performance in African contexts. Helm et al. (2024) link these technical gaps to broader social issues, arguing that biases in language technology can lead to epistemic injustice. Meanwhile, Adeyemi et al. (2023) propose innovative solutions like zero-shot cross-lingual reranking to improve model efficacy in low-resource settings. Alabi et al. (2020) compare massive and curated embeddings for languages such as Yorùbá and Twi, highlighting the enhanced effectiveness of tailored approaches over generic large-scale methods. Lastly, Dossou et al. (2023) focus on

adapting pretrained models for specific applications like low-resource clinical speech, utilizing epistemic uncertainty-based data selection to optimize performance. Collectively, these studies underscore the fact that current capabilities of LLMs to equitably represent global linguistic diversity are lacking. These studies also suggest targeted data enhancement and novel model adaptation techniques as viable pathways to bridge these gaps. However, the existing literature reveals notable gaps in connecting technical design choices with ethical considerations, often failing to test the latest technical solutions against an ethics “scorecard” of some sort. This is partly due to the fact that there is also a lack of continuity in the literature, with recent studies not building effectively on previous works, which results in isolated proposals that address specific issues but do not consider the broader implications or the strengths and weaknesses of other research.

A key opportunity here is to develop a framework for ethics based of Helm et. al. (2024) and then assess the technical solutions with this framework. Furthermore, while some studies may propose what appear to be best practices, there is a significant oversight in not examining how these practices may interact with each other for better or worse in terms of model performance, resource constraints, and ethics. This fragmentation highlights the need for developing a working set of best practices that promotes both model performance and ethics.

METHODS

First, we discussed and decided on the definition of ethicality in the scope of our project. The term "ethical," as defined in this context, refers to prioritizing normative diversity in models, ensuring they perform accurately to English-level standards, and focusing on language depth before

expanding to a breadth of languages. This approach prioritizes developing models that are principled and effective in specific linguistic and cultural contexts rather than those which are scalable.

Second, we identified 9 technical papers that were relevant to our research on building African Language NLP models. To be considered relevant, a paper had to either propose or critique a novel approach to developing these models, such as using different AI technologies, implementing unique data feeding methods like "zero-shot" learning, or addressing issues related to dataset size.

Then, we conducted a comprehensive literature review of 12 articles. Along with the 9 aforementioned technical papers, 3 of these 12 were non-technical, discussing the broader need for effective representation of African languages in NLP and associated ethical considerations.

Each paper was grouped into one of four categories, and each team member was assigned to review one paper from each group. After studying the papers, we gathered to share and discuss our findings. We created a matrix to capture key decisions or "variables" critiqued or made in each article, such as corpus size, and categorized these variables by scale (large, medium, or small).

As a group, we reached out the authors of each of the 12 articles for interviews to learn more about their stances on LLM ethicality. Of the 12 authors we contacted, 5 responded, and we conducted hour-long interviews over zoom with 2.

Finally, We analyze the ethicality, cultural sensitivity, and effectiveness of the identified

techniques to suggest directions for future research on language models that are both equitable and effective in African contexts. Reaching out to researchers in the field and conducting interviews have been substantial and informative for this project.

In order to organize our analysis of potential ethical LLMs, we deduced a series of key design decisions from our conversations with industry and field experts. For each of these design decisions, we identified potential courses of actions to evaluate the ethicality, cultural sensitivity, and effectiveness of. The design decisions and corresponding courses of action are described in Table 1.

Design Decision	Courses of Action
Modality	Multi, Dual, Single
Translation	Original Language, Translated to English
Tuning	Zero-Shot, Few-Shot/Language Adapter, Finetuning
Corpus Size	Large, Small

Table 1: Design Decisions and Corresponding Courses of Action to be Analyzed

Modality refers to modes of training data in the corpus used to train the LLM. Most production LLMs utilize a single-mode text training corpus. However, Mani and Namomsa (2023) mention discussing integrating images, sound bytes, and cultural artifacts into training data as a means of helping LLMs understand the local context of low-resource languages.

Translation refers to whether or not LLM queries written in low-resource languages are processes as-is in the original language, or translated to English first. Ojo et al. (2023) conducts preliminary analysis into the performance of pre-translated and original-language queries on GPT, LLaMa, and mT0 production LLMs.

Tuning refers to the process of how English-language LLMs are adapted to process African-language prompts. Adeyemi et al. (2023) describes the shortcomings of zero-shot learning, in which English-language LLMs are fed African-language prompts without alteration. Leong et al. (2023) discusses the challenges of finetuning models to process African-languages, as well as introduce high-performing few-shot techniques.

Corpus Size refers to the breadth of the training corpus used to train the LLM. While larger training corpuses provide more information for LLM training, they introduce noise into the model from training data not precisely curated to meet the model’s purpose. Dossou et al. (2023) introduces techniques for accurately training clinical LLMs on small but targeted African-Language training corpora.

Given the above design decisions and courses of action, we sought to find the best combination of courses of action based on expert interviews and literature reviews.

FINDINGS

Defining Ethicality

In order to consistently evaluate the ethicality of LLM design decisions, our team first had to

collect a series of metrics from which to define ethicality. As such, we interviewed Bonaventure Dossou, a McGill University PhD student researching Afrocentric NLP for healthcare contexts and Colin Leong, a Dayton University PhD student researching ML techniques for low-resource language Bible translation.

Dossou stressed that safety is the most important aspect of an ethical LLM. Specifically, an LLM should “embed values” and be “culturally aware,” such that it provides safe responses to user queries. Dossou provided the example of a translation app that is unaware of the local slang of region. If the translation app provides a culturally inappropriate greeting to a naive user, that user’s physical safety may be compromised when entering a new community. Dossou also emphasized that models must safely know when they are not confident about their answers. In the medical field, for example, if a tumor-detecting model asserts that a patient is tumor-free without a confidence level, that patient may not know what the correct path of treatment would be going forward.

Leong provided three metrics of ethicality that he had discussed with peer low-resource language researchers at the Missional AI conference—safety, credit, and usefulness. Leong stressed that a safe and useful NLP model is not ethical as long as the authors of the training set have not been credited or do not consent. Similarly, a safe NLP model is not ethical if it does not provide usefulness to its userbase, and a useful NLP model is not ethical if it is not safe.

Dossou and Leong’s interviews provided specific benchmarks of ethicality to supplement the more philosophical definitions discussed in the methods section (prioritizing normative diversity in models, ensuring they perform accurately to English-level standards, and focusing on language

depth before expanding to a breadth of languages).

Design Decision 1 — Modality:

Helm et al. (2024) argues that even if LLMs are trained on infinitely vast training corpora, they will still provide biased results as long as they incorporate anglo-centric architectural design decisions. One such design decision is the text-based modality of current LLMs. Mani and Namomsa (2023) emphasize that without visual training of models with artifacts, LLMs can never fully understand and reproduce a culture’s language. Especially for languages that rely on oral tradition or physical communication, a text-only training corpora will paper over many subtler forms of communication that cannot be conveyed through words. Thus, images and visual modes can help an LLM provide output that is culturally relevant to low-resource communities. Other modes of significance include temporality (to understand how the history of a community has shaped its communications and collective memory over time) and sensory inputs (such as descriptions of sounds, smells, and tastes common in a community).

We argue that incorporating multi-modal training data into an LLM corpus achieves ethicality for multiple reasons. First, multi-modal LLMs provide normative diversity to the NLP field. Instead of simply appending low-resource languages to training corpuses, multi-modal training data changes the fundamental assumptions of text-based LLM training data to provide a better representation of low-resource language communities. Second, multi-modal training data provides better performance to communities with oral or movement-based languages as it accounts for non-text aspects of language. As a consequence of better performance, these LLMs are also safer and more culturally sensitive/aware.

Design Decision 2 — Translation:

Most modern production LLMs have been trained primarily on English language corpuses. As such, African-language prompts passed into these models must either be pre-translated into English, or passed into the model as a zero-shot prompt.

Ojo et al. (2023) compared the performance of GPT-4, LLaMA 2, and mt0 LLMs on various NLP task prompts for nine different sub-Saharan African languages. While the models averaged a 0.2 F1 score for native African-language prompts, this score rose to 0.6 for prompts pre-translated to English. As such, we see that current LLM performance is much higher for pre-translated prompts as compared to native African language prompts.

Despite better performance on pre-translated prompts, we argue that native-language LLM querying is more ethical. First, we argue that pre-translation of African-language prompts to English is a form of “cosmetic diversity” — pretranslation is a quick way to adapt existing LLM infrastructure to African languages without investing significant time and resources into specifically creating LLMs for the African context. Although pre-translated queries can reach F1 scores of 0.6—much higher than native-language queries of 0.2—they are still far from the 0.96 F1 scores that these models have on native English-language queries. As such, pretranslation is a bandaid solution that circumvents deeper paradigm shifts that may better improve African-language LLMs.

Furthermore, machine-translation mechanisms for low-resource languages is lagging. As such, many of the cultural intricacies and meanings of African languages are lost upon translation to English. Thus, pre-translation risks the safety and

cultural authenticity of the resulting LLM response.

Given the ethical benefits of local-language queries, we advise future African-language LLMs to prioritize the intake of native-language queries.

Design Decision 3 — Tuning:

As mentioned in our discussion on pretranslation, zero-shot learning provides disappointing results when adapting English LLMs to African Languages (Average NLP Task F1 of 0.2). Finetuning production LLMs to African languages is promising, but can be computationally expensive. Leong et al. (2023) introduces a similar “low resource double bind” to that which we discussed in our introduction — the phenomenon that low-resource languages typically need high compute power to compensate for a lack of data, but regions with low-resource languages typically do not have high computer power available. Thus, although fine tuning may be helpful for many African languages, it is pragmatically infeasible to implement (and thus unethical to recommend).

Luckily, there exists middle ground between inaccurate zero-shot learning techniques and impractical finetuning techniques. Leong et al. (2023) investigates the use of “language adapters,” which are computationally-cheap tools for adapting production LLMs to local African languages. Such language adapters achieve similar performance on African languages as finetuned models do, yet are simple enough to be trained on a free Google Colab instance. Dossou et al. (2023) investigates the application of statistical methods of epistemic uncertainty towards reducing the amount of training data required to produce an accurate model.

In both of these cases, we see tuning solutions that are both accurate and lightweight, purpose-built for the African context. These innovations shift fundamental paradigms of LLM training in order to deliver quality models for African languages, proving commitment to normative diversity, performance, and cultural awareness.

Design Decision 4 — Corpus Size:

In the vast majority of LLM training applications, larger corpus sizes produce better results—GPT-4 leveraged 1.5 trillion parameters compared to GPT-3.5’s 175 billion, causing significant increases in performance. However, through expert interviews, we discovered that smaller corpus sizes may create more ethical African-Language LLMs.

First, smaller corpus sizes are safer. Dossou mentioned that chatbots created for clinical settings have few similarities with chatbots created for entertainment. As such, using an entertainment dataset to train the clinical chatbot may not provide many benefits to the model, but may have the adverse effect of introducing significant noise. Larger corpus sizes, though able to produce higher performing LLMs, also dilute the training corpus with documents unrelated to the LLM’s purpose. This may proliferate inaccurate, and potentially unsafe, information.

Second, larger corpus sizes run a larger risk of including unauthorized information. Leong described an ongoing study he is conducting about the sourcing of low-resource language datasets. Thus far, he has found 300 languages where the language’s only dataset has been compiled by Jehovah’s Witnesses. Upon realization of this fact, the Jehovah’s Witnesses promptly denied permission for future projects to use these datasets. While researchers like Leong may respect the wishes of dataset curators, it is

difficult to audit the information sources of larger datasets. Especially with massive, web-scraped datasets, information may come from sensitive, copyrighted, or sacred material. As such, attempting to enlarge corpus sizes may predispose the theft of intellectual property, which we consider unethical.

Considering the safety and intellectual property violations of larger datasets, we advise that ethical LLMs should focus on purposefully curating large yet specific datasets for model training.

Recommendations:

Our design choice recommendations for future ethical African-language ethical LLMs are summarized in Table 2.

Design Decision	Recommendation
Modality	Multi-modal training data
Translation	Native-language queries without pretranslation to English
Tuning	Computationally-cheap finetuning via language adapters and statistical optimization
Corpus Size	Smaller, purpose-specific training corpuses

Table 2: Recommended Design Decisions for Ethical LLMs

We hope that these design recommendations for African-Language LLMs will fill the existing literature gap about articulating the qualities of an “ethical” LLM.

DISCUSSION

Our research used findings from an extensive literature review and expert interviews to help address the urgent need for equitable representation and performance of African LLMs.

We found that the current approaches in language model development often overlook the unique cultural contexts of African languages, leading to suboptimal performances in real-world scenarios.

Here are some key areas where our research contributes to filling the gaps in the existing body of work:

Effectiveness of Model Tuning Techniques: Our study assessed various tuning strategies, including zero-shot learning, few-shot/language adapters and fine-tuning. We discovered that zero-shot learning, while cost effective, often fails to capture the nuanced semantic and syntactic properties of languages. In contrast, language adapters and fine-tuning, particularly when applied in a few-shot learning context, demonstrated superior performance. These findings suggest that investing in tailored strategies can significantly improve the functionality and accuracy of LLMs for African Languages, thus providing a direct response to the underperformance issues identified in prior research.

Cultural and Linguistic Representations: Through our interviews and literature analysis, we underscored the importance of integrating cultural and linguistic elements from the early stages of model development. Models trained on multi-modal data that included cultural artifacts (images, sound bytes) alongside textual data were

found to have a deeper understanding of the context and could generate more culturally relevant content. This approach not only enhances the quality of the generated language but also aids in preserving linguistic diversity and cultural heritage, addressing the cultural heritage, and addressing the cultural oversight noted in previous studies.

Optimal Corpus and Composition: Our findings challenge the prevailing assumption that larger datasets always result in better model performance. Instead, we identified a significant advantage in using curated, medium-sized datasets that are highly relevant to the specific tasks. This approach reduces noise and improves model accuracy, especially in clinical and educational applications where precision is critical. This is a very valuable insight as it addresses the gap in previous research that often neglected the impact of dataset composition and size on model performance in low-resource settings.

We encountered several challenges while completing our project. The first issue was a lower-than-expected response rate from experts. Although we planned to engage with at least eight experts, we only managed to receive responses from four, with just two answering our questions in virtual face-to-face meetings. Additionally, while it would have been ideal to design a prototype for an ethical African LLM based on the variables proposed in our project, constraints related to time, resources, and project scope made this unfeasible.

Addressing the constraints we identified would be a priority for potential future work. Our primary focus would be to engage more researchers in discussions about ethical considerations and gather their insights on this topic. We would also seek feedback on our work from researchers,

experts, and interested audiences, either by publishing our paper or directly sharing it with those who might find it most relevant. Finally, we would develop a prototype that adheres to the principles outlined in our research project.

CONTRIBUTION STATEMENT

Max planned and carried out the two expert interviews with the help of Nick-Aurel. Max analyzed and structured the findings of the study. Nick-Aurel contributed to the project by brainstorming ideas, interviewing experts with Max, and writing the Methods, Related Work, and Discussion sections with assistance from Kofi. Kofi wrote the abstract, introduction, and assisted Nick-Aurel in writing the related work for this paper. Obioha contributed to defining the project proposal, assisting Kofi and Nick-Aurel with the abstract, introduction, and related work, and proofreading the paper, as well as writing references.

REFERENCES

- [1] Mofetoluwa Adeyemi, Akintunde Oladipo, Ronak Pradeep, Jimmy Lin. 2023. Zero-Shot Cross-Lingual Reranking with Large Language Models for Low-Resource Languages. arXiv:2312.16159v1
- [2] Jesujoba O. Alabi, Kwabena Amponsah-Kaakyire, David I. Adelani, Cristina Espana-Bonet. 2020. Massive vs. Curated Embeddings for Low-Resourced Languages: the Case of Yorub`a and Twi. In Proceedings of the 12th Conference on Language Resources and Evaluation (LREC 2020), pages 2754–2762 Marseille, 11–16 May 2020.
- [3] Eleni Coundouriotis. 2022. AUGMENTING THE DATA: VUKOSI MARIVATE ON AFRICAN NATURAL LANGUAGE PROCESSING (NLP). Retrieved February 28, 2024 from <https://criticalai.org/2023/01/24/augmenting-the-data-vukosi-marivate-on-african-natural-language-processing-nlp/>
- [4] Bonaventure F. P. Dossou et. al. 2023. Adapting Pretrained ASR Models to Low-resource Clinical Speech using Epistemic Uncertainty-based Data Selection. arXiv:2306.02105v2
- [5] Bonaventure F. P. Dossou, Iffanice Houndayi, Pameley Zantou, Gilles Hacheme. 2023. FonMTL: Towards Multitask Learning for the Fon Language. arXiv:2308.14280v2
- [6] Paula Helm, Gábor Bella, Gertraud Koch, Fausto Giunchiglia. 2024. Diversity and language technology: How language modeling bias causes epistemic injustice. <https://doi.org/10.1007/s10676-023-09742-6>
- [7] Kreutzera et. al. 2022. Quality at a Glance: An Audit of Web-Crawled Multilingual Datasets. arXiv:2103.12028v4
- [8] Ganesh Mani, Galane Basha Namomsa. 2023. IEEE AFRICON. Large Language Models (LLMs): Representation Matters, Low-resource Languages and Multi-Modal Architecture. Retrieved February 29, 2024 from IEEE Xplore.
- [9] Leong et. al. 2023, Adapting to the Low-Resource Double-Bind: Investigating Low-Compute Methods on Low-Resource African Languages. AfricaNLP workshop at ICLR2023. arXiv:2303.16985v1
- [10] Rohin Manvi, Samar Khanna, Marshall Burke, David Lobell, Stefano Ermon. 2024. Large Language Models are Geographically Biased. Retrieved February 28, 2024.
- [11] Olatunji et. al. 2023. AfriSpeech-200: Pan-African Accented Speech Dataset for Clinical and General Domain ASR. arXiv:2310.00274v1
- [12] Jessica Ojo, Kelechi Ogueji, Pontus Stenetorp, David Ifeoluwa Adelani. 2023. How good are Large Lanugage Models on

African Languages? Retrieved February 29, 2024.

- [13] Zindi. (July 15, 2024) Zindi and AI4D build language datasets for African NLP. Retrieved February 29, 2024 from <https://zindi.medium.com/zindi-and-ai4d-build-language-datasets-for-african-nlp-34a4d0ea129>